Additional Reference for Assignment 2

October 25, 2023

In large sample asymptotic theory, we need the following definition,
Definition [Convergence in Probability] A sequence of random variables $X_1, X_2, X_3, \cdots$ converges in probability to a random variable X, shown by $X_n \xrightarrow{p} X$, if

$$\lim_{n\to\infty} P\left(|X_n - X| \geq \epsilon\right) = 0, \quad \text{for all } \epsilon > 0.$$

Explanation For Question 1 (2)
For the expression of the omitted variable bias, recall in SLR, the regressor is

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)u_i}{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2}$$

and note that under the SLR assumptions,

$$\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \xrightarrow{p} \text{var}\left(X_i\right)$$

and

$$\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)u_i \xrightarrow{p} \text{cov}\left(X_i, u_i\right)$$

since $\mathbb{E}[u] = 0$.

And note also that - glossing over some mathematical detail - substituting these probability limits into the regressor expression yields

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \frac{\text{cov}\left(X_i, u_i\right)}{\text{var}\left(X_i\right)}$$

Expressing $\text{cov}\left(X_i, u_i\right)$ and $\text{var}\left(X_i\right)$ in terms of the correlation coefficient between X and u, ρ_{Xu}, the standard deviation of X, σ_X, and standard deviation of u, σ_u. We can show that

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \rho_{Xu}\frac{\sigma_u}{\sigma_X}$$